MahimandalSR: A Same-Day, Block-Disjoint Landsat–Sentinel-2 RGB–NIR Dataset for Real 3× Super-Resolution Across Bharat

September 2026

Abstract

Real satellite-image super-resolution requires paired observations that are close in time, geometrically compatible, radiometrically interpretable, and geographically diverse. We introduce MahimandalSR, a dataset of 4,800 georeferenced Landsat 8/9 Operational Land Imager and Sentinel-2 MultiSpectral Instrument surface-reflectance pairs spanning Bharat. Each pair represents the same nominal 4.8 km × 4.8 km footprint as a four-band blue–green–red–near-infrared Landsat input at 30 m (160 × 160 pixels) and Sentinel-2 reference target at 10 m (480 × 480 pixels), not exact 10-m Landsat ground truth. Within every pair, the two observations share a UTC acquisition date; the mean time separation is 8.89 minutes and the maximum is 19.8 minutes. Patch-level joint cloud and shadow obstruction averages 0.123%. Sampling combines the Esri/Impact Observatory 2025 annual 10-m land-cover product with Copernicus DEM GLO-30 to represent nine landscape strata, with a dedicated western-Himalayan collection phase contributing 828 pairs. Complete 100-km spatial blocks are assigned to one split, without a guaranteed intersplit footprint gap. The audited manifest contains 3,424 training pairs, 645 validation pairs, and 731 test pairs from 13 held-out blocks; source products can recur across blocks. A gradient phase-correlation audit yields a mean residual displacement proxy of 0.077 Sentinel-2 pixels and a 95th percentile of 0.160 pixels; this is an image-based consistency indicator, not surveyed ground-control-point accuracy. On the 731-pair reflectance16 test split, RDN reaches 37.67 dB RGB PSNR and 36.15 dB four-band PSNR, improving over bicubic by 2.29 and 2.26 dB, respectively. These single-seed benchmarks establish a reference, not a claim of model superiority beyond this release. The dataset retains geospatial metadata, 16-bit GeoTIFFs, per-pair quality fields, and reproducible reports.

Keywords: satellite image super-resolution; Landsat; Sentinel-2; cross-sensor dataset; RGB-NIR; Bharat; spatial leakage; surface reflectance

1 Introduction

Landsat provides a long and radiometrically calibrated record of land-surface change, but its commonly used reflective bands have a 30-m ground sampling distance. Sentinel-2 provides corresponding visible and near-infrared bands at 10 m, creating a practically important 3× cross-sensor super-resolution setting. Unlike synthetic single-image super-resolution, however, a Landsat–Sentinel pair is not generated by applying a known blur and downsampling operator to one image. The sensors differ in spectral response, acquisition geometry, atmospheric correction, point-spread function, and observation time. Consequently, dataset construction is part of the scientific problem rather than a preprocessing detail (Claverie et al., 2018; Wang et al., 2021a).

OLI2MSI established a real multisensor benchmark using Landsat-8 inputs and Sentinel-2 references, with 5,225 training and 100 test pairs (Wang et al., 2021a,b). Land2Sent subsequently provided 15,066 four-band tiles derived from 30 manually selected 2023 source pairs and used acquisition separations below one hour (Wang et al., 2026; RSandAI, 2026). These resources demonstrate the value of real cross-sensor pairing. They also motivate complementary datasets that expose regional diversity, make patch-level quality controls explicit, and evaluate the effect of grouping samples geographically.

MahimandalSR is designed around those requirements for Bharat. Its contribution is not a new super-resolution architecture. It is a documented data resource whose construction emphasizes same-day and near-simultaneous matching, cloud and shadow control, sampled-location uniqueness, terrain and land-cover diversity, and whole-block partitioning. The principal contributions are:

- 4,800 real RGB-NIR surface-reflectance pairs at the standard $160 \times 160 \rightarrow 480 \times 480$ geometry, covering 298 Landsat scenes, 648 Sentinel-2 products, and 311 Sentinel-2 MGRS tiles;
- a strict same-UTC-date protocol with a measured 0.1–19.8 minute acquisition-time separation and explicit patch-level quality filters;
- transparent, metadata-driven sampling across nine landscape strata, including a western-Himalayan reserve and terrain attributes from Copernicus DEM GLO-30;
- spatial-block partitioning, pair-level provenance, physical-reflectance storage, and an audit bundle that records selection, rejection, composition, and consistency statistics;
- a single-seed reflectance16 baseline benchmark on the held-out 731 pairs, with RGB and four-band results, spectral-index error, classwise breakdowns, and source-implementation qualifications.

2 Related Datasets and Design Context

Synthetic super-resolution datasets obtain LR inputs by degrading a high-resolution image. This gives perfectly aligned pairs but does not reproduce the sensor and atmospheric differences encountered in operational Landsat enhancement. OLI2MSI instead pairs Landsat-8 OLI and Sentinel-2 MSI observations at 30 m and 10 m, respectively (Wang et al., 2021a). Its public release contains full 160×160 LR and 480×480 HR GeoTIFFs, but its commonly used split is location-random and its published metadata do not expose the strata-aware, patch-level audit fields used here (Wang et al., 2021b).

Land2Sent uses the same $3\times$ geometry and four RGB-NIR bands. It selected 30 cloudless 2023 source pairs with less than one hour between acquisitions, then tiled them into 15,066 samples with a 70/20/10 split (RSandAI, 2026; Wang et al., 2026). MahimandalSR follows the compatible geometry and band choice so models can be transferred without architectural modification, but differs in four ways: (i) every released pair is same-day and within 19.8 minutes in the audited manifest; (ii) patch selection is automated and records every source product and quality decision; (iii) sampling explicitly reserves Himalayan and less common landscape regimes; and (iv) splits operate on 100-km spatial blocks rather than independent patches. This prevents identical block IDs from entering multiple splits but does not establish a 100-km gap, or even a nonzero footprint gap, between partitions (Roberts et al., 2017).

Table 1 separates directly reported facts from unavailable comparisons. "NR" means that a numerically comparable statistic was not reported in the cited public paper or repository; it does not imply inferior quality. In particular, the three datasets use different selection and split protocols, so their aggregate statistics should not be treated as a leaderboard.

The term "ground truth" is used cautiously throughout this paper. Sentinel-2 is the higher-resolution reference, but it is not a radiometrically exact observation of what Landsat would have measured at 10 m. Spectral-response and point-spread-function differences remain part of the mapping task (Claverie et al., 2018). We therefore refer to the Sentinel-2 image as the *target* or *reference*.

3 Dataset Construction

3.1 Source imagery and catalogue search

Landsat Collection 2 Level-2 and Sentinel-2 Level-2A products were queried through the Microsoft Planetary Computer STAC API (Microsoft, 2026). Landsat candidates were restricted to Landsat 8 and 9

Table 1: Quantitative protocol comparison using public dataset papers and repositories. Timing is the source-pair acquisition criterion or measured released-pair distribution. Registration and spectral values are reported only when computed under a stated comparable audit.

Dataset	Pairs	LR/HR	Bands	Train/validation/test
OLI2MSI (Wang et al., 2021a,b)	5,325	160/480	RGB	5,225/–/100
Land2Sent (Wang et al., 2026; RSandAI, 2026)	15,066	160/480	RGB-NIR	70/20/10%
MahimandalSR	4,800	160/480	RGB-NIR	71.33/13.44/15.23%

Dataset	Acquisition timing	Residual registration	Downsampled spectral agreement
OLI2MSI	NR	NR	NR
Land2Sent	< 60 min at source-pair selection; distribution NR	NR	NR
MahimandalSR	100% same date; mean 8.89 min; max 19.8 min	Mean 0.0767 px; P95 0.1598 px [a]	Pearson B/G/R/NIR: 0.862/0.914/0.950/0.925

[a] Gradient phase-correlation residual proxy on the Sentinel-2 grid, not surveyed geodetic accuracy. OLI2MSI provides a random pair split; MahimandalSR uses indivisible 100-km spatial blocks. Land2Sent reports a 70/20/10 split in its public documentation.

Table 2: Bands retained in each pair. Wavelength intervals are approximate sensor specifications recorded by the preparation workflow.

Channel	Landsat 8/9 OLI asset	Approx. μm	Sentinel-2 MSI asset
Blue	blue	0.45–0.51	B02 (0.46–0.52)
Green	green	0.53–0.59	B03 (0.54–0.58)
Red	red	0.64–0.67	B04 (0.65–0.68)
NIR	nir08	0.85–0.88	B08 (0.78–0.90)

Tier-1 products; Sentinel candidates included Sentinel-2A, 2B, and 2C. Landsat Collection 2 Level-2 provides atmospherically corrected surface-reflectance products (U.S. Geological Survey, 2021). Sentinel-2 MSI has four visible and near-infrared bands at 10 m, among its 13 spectral bands (European Space Agency, 2026b). We retain RGB-NIR because these are the four common bands natively available at the required Landsat 30-m and Sentinel-2 10-m resolutions and because this scope is compatible with OLI2MSI/Land2Sent studies. Sentinel-2 SWIR is native at 20 m rather than 10 m; including it would introduce an additional resampling protocol instead of the same 3× sensor mapping. Table 2 gives the paired channels.

The search interval represented in the final manifest is 9 March 2025 to 11 September 2026. Candidate Landsat and Sentinel products had to intersect inside the Bharat boundary used by the collection notebook and share the same UTC date. The configured maximum separation was 180 minutes, although every retained pair is separated by at most 19.8 minutes. The collection proceeded in two passes: a western-Himalayan priority pass over Jammu and Kashmir, Ladakh, Himachal Pradesh, and Uttarakhand, followed by a balanced Bharat-wide pass. Administrative labels were assigned with geoBoundaries gbOpen ADM1 data (Runfola et al., 2020). This boundary source is reproducible and openly licensed, but it is not an official Survey of India boundary and should not be used to infer a political position. Maps in this article show sample centres and analysis grids, not international boundaries or a legally authoritative national boundary. Their depiction and labels do not express a position on sovereignty, the Line of Control, or the final status or delimitation of any territory.

3.2 Patch geometry, gridding, and reflectance

Each Landsat patch is an aligned 160×160 window on the native 30-m projected grid. Its bounds define a 480×480 Sentinel-2 target grid at 10 m, yielding the same nominal 4.8 km square footprint. Sentinel reflectance bands are warped to this target grid with bilinear interpolation; Sentinel scene-classification labels use nearest-neighbour resampling. The resulting LR and HR rasters share a coordinate reference system and equal projected bounds.

The preparation code converts Landsat Collection 2 Level-2 digital numbers as

$$r_L = \mathrm{clip}(0.275\, DN_L - 2000,\, 0,\, 10000), \tag{1}$$

and subtracts the Sentinel-2 baseline offset of 1,000 for processing baselines at or above 04.00 before applying the same $[0, 10000]$ bound. Values are rounded and stored as signed 16-bit reflectance multiplied by 10,000. Statistical analysis converts them to physical reflectance and clips to $[0, 1]$. This physical clipping is separate from visualization: figures use a fixed $[0, 0.30]$ reflectance display stretch and a stated display-only gamma.

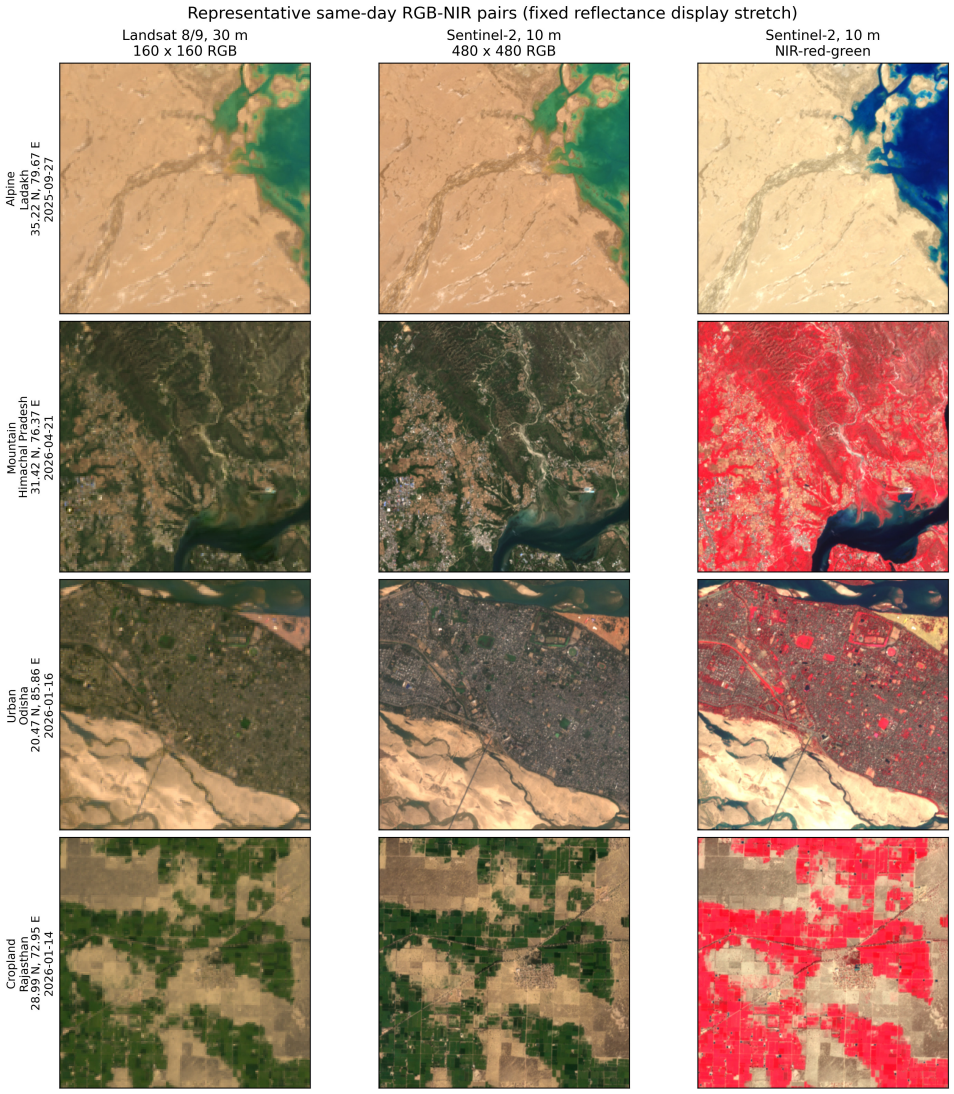


Figure 1: Representative manifest-traceable pairs. Each row shows the same nominal footprint as a Landsat RGB input, Sentinel-2 RGB target, and Sentinel-2 NIR–red–green composite. The fixed $[0, 0.30]$ reflectance stretch and display-only gamma are shared across panels. Coordinates, acquisition dates, and pair identifiers are recorded in the accompanying provenance CSV.

3.3 Cloud, no-data, texture, and agreement filtering

The completed original-source audit reproduces the stored red-band conversion without pixel mismatches for three checked Landsat–Sentinel pairs. Separately, an independent product-metadata audit retrieves and hashes the original XML for all 648 Sentinel products representing the 4,800 pairs. Every product specifies a BOA quantification value of 10,000 and additive offsets of -1000 DN for B02, B03, B04, and B08, agreeing with the preparation formula. Product identifiers, physical-band mappings, and processing baselines are checked against STAC; 323 products use baseline 05.11 and 325 use 05.12. This verifies the independently recorded conversion coefficients, rather than inferring physical scaling from default raster-reader metadata (Copernicus Programme, 2026). It does not replace an all-pixel source audit or prove provider-asset harmonization (Supplementary S3). No release encoding or trained checkpoint is changed by this check.

A subsequent reader-level check reconstructs all four bands of two deterministically selected source pairs: all 2,048,000 stored band values agree with the release. The readers return integer DN without

implicitly applying the physical calibration; a double-offset control disagrees with every pixel in the eight checked Sentinel windows. This sampled check is supplemented by an original-source audit of all 1,187 rasters containing stored zeros, described in Supplementary S9. Stored zero is not itself a validity label: valid dark reflectance can clip to the same value as source fill. The collector checks raw zero DN before clipping but permits up to 0.1% such locations and does not export a separate validity mask into the original TIFFs.

Scene-level cloud metadata were used only as a broad catalogue prefilter (at most 60% for either source); all strict decisions were made on the candidate patch. A Landsat pixel was considered clear when the five least-significant cloud-related QA bits were zero. Sentinel pixels were considered clear for Scene Classification Layer classes vegetation, bare soil, water, and snow/ice. At least 99% of the upsampled Landsat and Sentinel joint mask had to be clear. Patches with more than 50% Sentinel water or more than 0.1% zero-valued pixels in either source were rejected.

To detect gross mismatch, the Sentinel target was reduced to the Landsat grid by averaging non-overlapping 3×3 blocks. Pearson correlation was then measured against Landsat red and NIR. The normal-terrain threshold was 0.80 in both bands; alpine patches used 0.65 to accommodate snow, shadow, and steep relief. The Sentinel NIR standard deviation had to exceed 150 stored reflectance units, relaxed to 40 for alpine terrain. These thresholds are selection diagnostics, not claims of sensor equivalence.

3.4 Land-cover and terrain-aware sampling

The provenance audit retains the exact preparation-code classification rules, Esri service URL and 2025 year filter, and current Esri and Copernicus DEM catalogue responses with retrieval timestamps and hashes. These current responses are not historical snapshots of the original requests. Per-patch contributing Esri raster IDs and DEM item/version receipts were not established by this audit, so exact historical auxiliary-raster lineage remains partial.

The 2025 Esri/Impact Observatory Sentinel-2 10-m annual land-use/land-cover map supplied patch composition for water, trees, flooded vegetation, crops, built area, bare ground, snow/ice, clouds, and rangeland (Impact Observatory et al., 2026). Copernicus DEM GLO-30 supplied elevation and slope descriptors; GLO-30 is a digital surface model rather than a bare-earth terrain model (European Space Agency, 2026a). ESA WorldCover 2021 v200 was retained only as a historical comparison field, not as the release's contemporaneous label source (Zanaga et al., 2022).

One mutually exclusive sampling stratum was assigned with transparent rules. Alpine terrain was identified by at least 1% snow/ice, 90th-percentile elevation at least 3,000 m, or the combination of elevation at least 2,500 m and 90th-percentile slope at least 15°. Mountain preceded the land-cover rules when 90th-percentile elevation was at least 1,500 m or slope at least 15°. The remaining rules used dominant built, bare, tree, crop, water/flooded-vegetation, and rangeland percentages; unresolved mixtures entered the mixed class. Hard class caps prevented cropland and other frequently discovered classes from consuming the release. Candidate locations overlapping a previously accepted footprint by 50% or more were rejected before expensive raster reads.

3.5 Spatial uniqueness and split assignment

Accepted patch centres were projected to an equal-area coordinate system. Overlapping samples were grouped and the highest-quality pair in each group was retained. The executed release configuration used `guard_m=0`: no boundary guard was applied. Release construction retained 4,800 unique centres and grouped them into 274 spatial blocks of 100 km. Block identifiers were calculated by taking the floor of each equal-area coordinate divided by the block size. Complete blocks were assigned to train, validation, or test, so no block appears in more than one split. This rule does not prevent footprints near adjacent block boundaries from overlapping across partitions.

The release allocator greedily balanced landscape class, geographic region, and priority region while keeping complete blocks intact. Assignment accounts for missing categories, desired split size, and an

overfilling penalty, with deterministic seed-based tie breaking. Validation and test were each required to contain at least 11% of pairs; complete blocks were moved from training when needed. Because block populations differ and blocks remain indivisible, the realized fractions exceed these minima and the two holdouts need not have equal counts.

The resulting manifest contains 3,424 training pairs (71.33%) from 247 blocks, 645 validation pairs (13.44%) from 14 blocks, and 731 test pairs (15.23%) from 13 blocks. These are spatially grouped, category-aware partitions rather than an exact random image-level split or equal-sized landscape classes. Table 3 reports the realized percentages. Any future allocation change requires a newly versioned manifest and regenerated reports.

Spatial-block disjointness does not imply source-product disjointness. An audit of the frozen reflectance16 release manifest (SHA-256 `6bf97403ee40ccf0`; full digest in the supplementary audit) found zero blocks shared between partitions, but 55 Landsat and 64 Sentinel-2 IDs shared between training and test; 56 and 64, respectively, are shared between training and validation. Training and test share 68 exact source-product combinations, represented by 381 test patches. The completed geometry audit transformed raster-header footprint polygons to EPSG:6933 and measured positive-area intersections using a $> 1\,\mathrm{m}^2$ numerical threshold. Fourteen test patches intersect training footprints and nine intersect validation footprints, with one in both groups: 22 unique test patches are affected. Minimum cross-split footprint distances are zero for every split comparison. These projected-footprint diagnostics are not surveyed geodetic accuracy measurements. The benchmark therefore tests held-out *block IDs*, not guaranteed spatially buffered locations or wholly unseen acquisitions. We retain the frozen benchmark and additionally report a 709-pair overlap-excluded sensitivity analysis; a source-disjoint benchmark would require a new split and retraining. Supplementary Sections S1, S3, and S6 document these distinctions.

4 Data Records and Dataset Characteristics

4.1 Release structure

For each pair identifier, the release contains one four-band Landsat GeoTIFF and one four-band Sentinel-2 GeoTIFF. Files preserve CRS, affine transform, bounds, band descriptions, source product identifiers, landscape class, and administrative label. The manifest records source IDs, acquisition times, cloud metadata, centre coordinates, equal-area coordinates, reflectance statistics, land-cover percentages, terrain summaries, residual-shift diagnostics, split block, and quality flag. Separate `train.csv`, `val.csv`, and `test.csv` files expose the partition without moving imagery. The manifest-defined imagery occupies 6.464 GiB: 0.666 GiB for Landsat and 5.798 GiB for Sentinel-2.

An independent file audit opened all 9,600 manifest-referenced rasters. Every pair had four bands ordered blue, green, red, NIR; the expected 160×160 and 480×480 dimensions; 30-m and 10-m pixel spacing; signed 16-bit storage; equal LR/HR projected bounds; and matching CRS. No manifest-referenced file was missing or malformed. The raw staging directory contained three complete LR/HR pairs that were not selected into the frozen manifest. Manifest-driven packaging excluded those staging artifacts. Both release variants contain exactly 9,600 TIFFs referenced by 4,800 manifest rows, with zero missing and zero orphan TIFFs; the exporter now treats any closure mismatch as a fatal error.

4.2 Geographic and thematic composition

Patch centres extend from 8.397°N to 36.430°N and 68.485°E to 95.827°E. The manifest includes 28 named state or union-territory labels plus two records whose administrative label is unavailable. The dedicated Himalayan phase produced 334 pairs in Ladakh, 275 in Jammu and Kashmir, 134 in Himachal Pradesh, and 85 in Uttarakhand. Figure 2 shows broad national coverage and the spatially disjoint split blocks.

The class caps limited cropland to 20.83% and produced 779–1,000 examples in five common strata. The initial reserve targets were not achieved for desert/bare (50 obtained versus 450 requested), wa-

Table 3: Audited release overview. Percentages are relative to 4,800 manifest pairs.

Quantity	Count	Percent
Training split	3,424	71.33%
Validation split	645	13.44%
Test split	731	15.23%
Test 100-km spatial blocks	13	–
Landsat 8 / Landsat 9	2,719 / 2,081	56.65 / 43.35%
Excellent / good / acceptable quality	3,037 / 1,613 / 150	63.27 / 33.60 / 3.13%
Western-Himalayan priority region	828	17.25%
2025 / 2026 acquisitions	549 / 4,251	11.44 / 88.56%
Unique 100-km spatial blocks	274	–

Validation and test each had an 11% minimum; whole-block assignment produced the realized train/validation/test percentages of 71.33/13.44/15.23. Excellent means clear fraction ≥ 0.995 and minimum red/NIR correlation ≥ 0.90; good means clear fraction ≥ 0.99 and minimum correlation ≥ 0.80; all other retained alpine-threshold cases are acceptable.

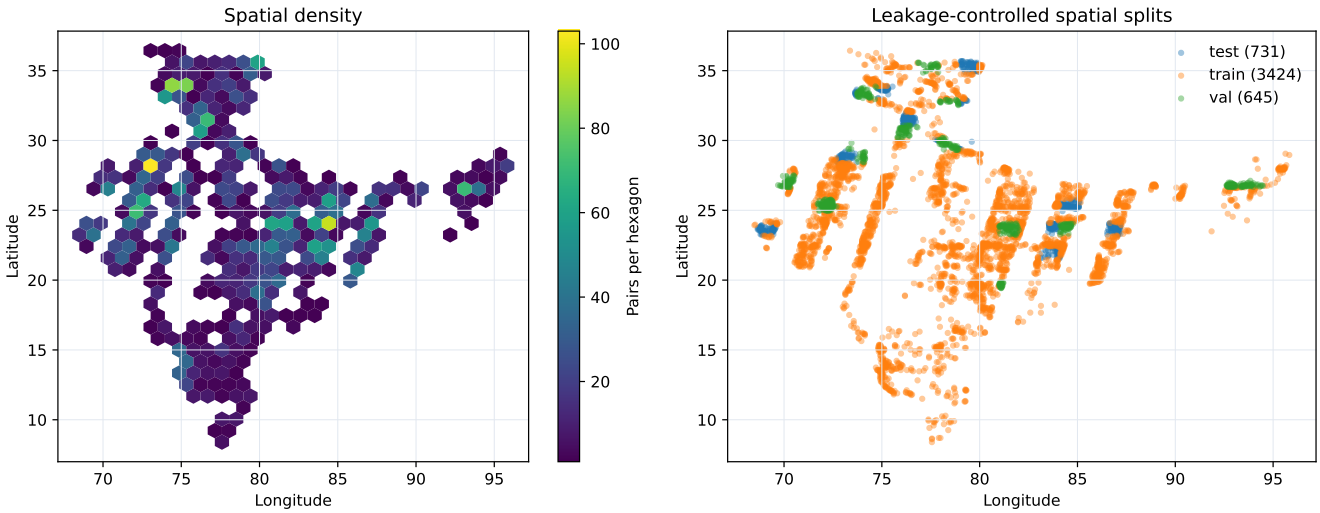


Figure 2: Spatial density (left) and the current block-disjoint train/validation/test assignment (right). Dense sampling reflects catalogue availability and quality acceptance, not population or land area. Points and analysis grids do not depict international boundaries or imply a position on sovereignty, the Line of Control, or final territorial status.

Table 4: Landscape sampling strata in the 4,800-pair manifest. Strata are sampling metadata derived from annual land cover and DEM thresholds, not manual semantic ground truth.

Stratum	Pairs	Percent	Stratum	Pairs	Percent
Cropland	1,000	20.83	Alpine	353	7.35
Urban	800	16.67	Mixed	162	3.38
Mountain	800	16.67	Water/wetland	56	1.17
Shrub/grass	800	16.67	Desert/bare	50	1.04
Forest	779	16.23			

ter/wetland (56 versus 300), and mixed (162 versus 500). This shortfall is reported rather than backfilled with low-quality or duplicate patches. The release is strata-aware, not class-uniform: cropland outnumbers desert/bare by 20:1. All nine strata occur in each split, but the five desert/bare test pairs come from

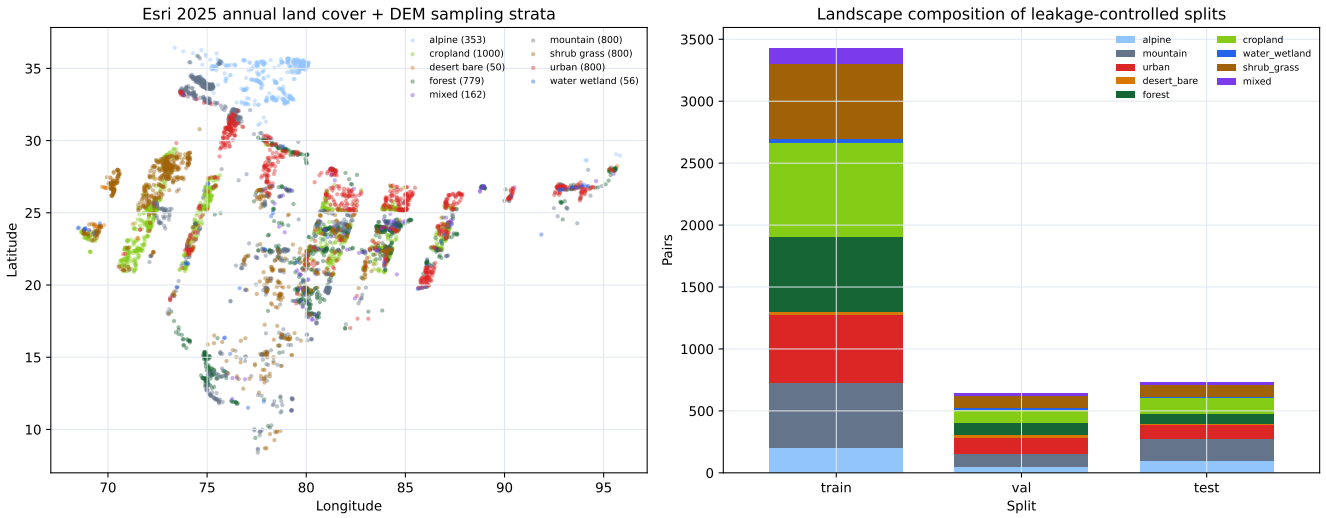


Figure 3: Sampling strata across Bharat (left) and their counts in each current split (right). Landscape labels combine Esri 2025 annual land-cover percentages with Copernicus DEM thresholds. Points do not depict international boundaries or imply a position on sovereignty, the Line of Control, or final territorial status.

one block and the seven water/wetland test pairs from two. Supplementary Table S3 gives the exact pair and distinct-block counts; Figure 3 visualizes the distribution. These labels describe patch-level sampling strata, not per-pixel semantic ground truth.

Terrain varies from low-lying coastal and riverine regions to Himalayan surfaces. Mean and median patch elevation are 642.6 m and 262.6 m, the 90th percentile is 1,647.4 m, and the maximum is 7,235.8 m. Mean slope is 4.13° and its 90th percentile is 20.57°. The annual Esri labels indicate mean patch compositions of 37.18% crops, 26.64% rangeland, 19.33% trees, 8.52% built area, 5.32% bare ground, 2.40% water, and smaller flooded-vegetation, snow/ice, and cloud fractions.

4.3 Temporal and atmospheric characteristics

For each of the 4,800 pairs, Landsat and Sentinel-2 share a UTC acquisition date; pairs span 9 March 2025 through 11 September 2026. Within-pair acquisition-time separation averages 8.895 minutes, with a 9.2-minute median, 15.3-minute 95th percentile, and 19.8-minute maximum. This is materially stricter than a same-day-only statement and reduces, but does not eliminate, moving-cloud, illumination, water, and human-activity differences.

The dataset is temporally imbalanced. It contains 549 pairs from 2025 and 4,251 from 2026. January 2026 alone contributes 2,555 pairs; October contributes none and November only two. The imbalance follows the conjunction of source availability, same-day orbital opportunities, cloud filtering, geographic reserves, and the stopping rule at 4,800 accepted pairs. Supplementary Table S4 cross-tabulates acquisition month, year, and broad sampling region. Seasonal comparisons therefore require reweighting or stratified subsampling rather than treating the release as a climatological sample.

Re-reading the original Landsat QA and Sentinel-2 SCL masks succeeded for all 4,800 pairs. Unrounded joint cloud and shadow obstruction averages 0.12286% and reaches a maximum of 0.99740%; every pair satisfies the below-1% obstruction rule. The three manifest fractions rounded to 0.9900 have unrounded clear fractions of 0.99003472, 0.99003472, and 0.99002604, resolving the apparent boundary ambiguity. Source-scene metadata average 5.27% cloud for Landsat and 5.56% for Sentinel-2, illustrating why patch-level masks are more informative than scene-level cloud percentages.

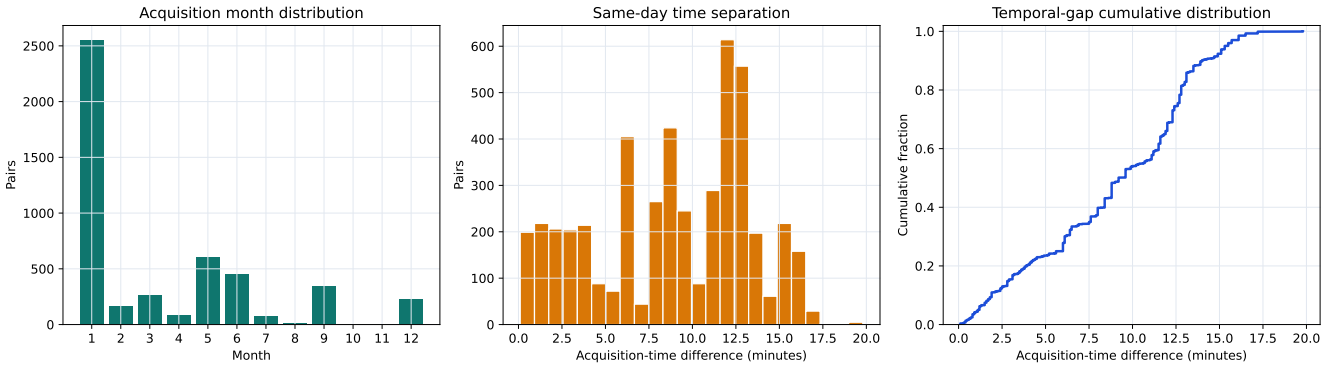


Figure 4: Acquisition-month distribution, same-day time separation, and its cumulative distribution. The strong month imbalance is an observed property of the release and should be considered in seasonal analyses.

4.4 Geometric and radiometric consistency

Geometric compatibility was audited after resampling using gradient-magnitude phase correlation with subpixel refinement (Guizar-Sicairos et al., 2008). The mean residual-shift proxy is 0.0767 Sentinel-2 pixels (0.767 m), the median is 0.0678 pixels, the 95th percentile is 0.1598 pixels (1.598 m), and the maximum is 0.6242 pixels. These values summarize image-to-image residual consistency. They are not independent geodetic accuracy measurements and must not be reported as ground-control-point RMSE.

A second, CPU-only diagnostic uses a fixed class-stratified sample of 100 pairs from 90 spatial blocks and 99 distinct acquisition pairs. Sentinel-2 is area-reduced to the 30-m Landsat grid. SIFT descriptor matches (Lowe, 2004) are partitioned by source-image spatial cells before a similarity transform is fitted on training matches only. All held-out matches are then scored without residual filtering. Ninety-eight pairs are evaluable; two alpine pairs lack enough spatially held-out matches. Across the 98 evaluable pairs, the mean per-pair median displacement is 0.253 Landsat pixels (0.759 target pixels), while the mean per-pair RMS displacement is 0.692 Landsat pixels (2.076 target pixels). After the diagnostic transform, mean held-out residual RMS is 0.677 Landsat pixels. The larger RMS values retain gross descriptor mismatches; they are not discarded to meet a registration threshold. Neither these automatic matches nor the original phase proxy are surveyed control points. No release image is corrected by this diagnostic. Supplementary S7 reports the protocol, failures, classwise results, and inspection evidence.

After reducing Sentinel-2 to the Landsat grid, the mean cross-sensor reflectance MAE is 0.0160 and RMSE is 0.0210. Mean Pearson correlations are 0.862 for blue, 0.914 for green, 0.950 for red, and 0.925 for NIR. Landsat mean reflectance is 0.0673, 0.1035, 0.1130, and 0.2540 for blue through NIR; Sentinel-2 means are 0.0812, 0.1101, 0.1237, and 0.2603. The nonzero offsets reinforce that the target differs radiometrically from a hypothetical 10-m Landsat observation.

Table 5: Cross-sensor agreement after Sentinel-2 is reduced to the Landsat grid.

Metric	Blue	Green	Red	NIR
Pearson correlation	0.8620	0.9141	0.9502	0.9251
Landsat mean reflectance	0.0673	0.1035	0.1130	0.2540
Sentinel-2 mean reflectance	0.0812	0.1101	0.1237	0.2603

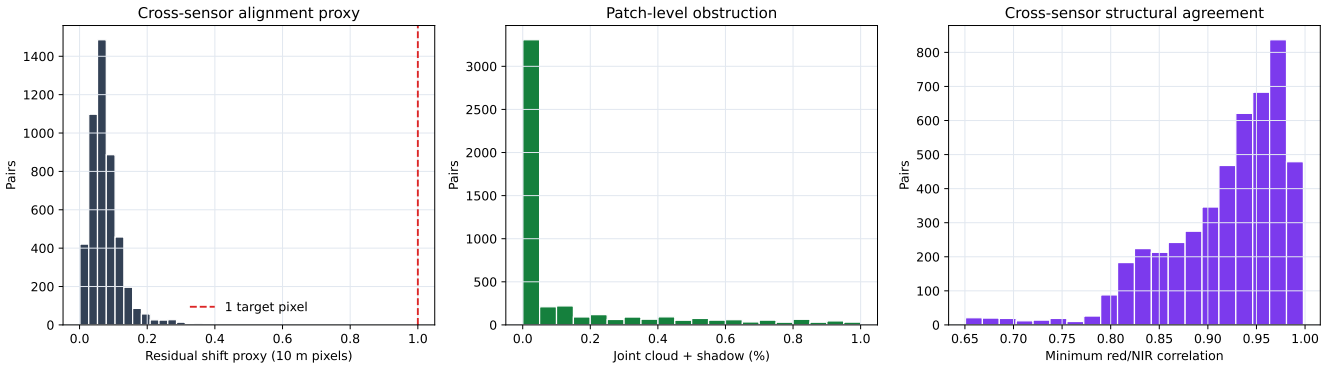


Figure 5: Release quality distributions. The red reference line in the left panel denotes one Sentinel-2 target pixel. Registration values are phase-correlation residual proxies, not surveyed positional error.

5 Benchmark Experiments

5.1 Protocol and model coverage

We evaluated the frozen reflectance16 release on its block-disjoint, source-product-overlapping test split of 731 patches from 13 held-out 100-km blocks. The completed audit finds positive-area cross-split footprint intersections, so this is not a spatially buffered test. All trainable models used the same 3,424 training and 645 validation pairs; bicubic required no training. Training consumed the full 160×160 Landsat image and its aligned 480×480 Sentinel-2 target (batch size 1), with aligned dihedral augmentation, masked L1 loss, Adam at 10^{-4}, a 50-epoch ceiling, and early stopping after three validation epochs without improvement. The best full-frame validation-PSNR checkpoint was evaluated once on all 731 test pairs. This is one training seed (42), not a multi-seed estimate. Only validated author or explicitly identified comparison implementations that passed the input/output and backward-pass preflight were run; source revisions and architecture settings are recorded in per-model protocol files and summarized in Supplementary Table S5. The RGB panel comprises EDSR, RCAN, RDN, CAFRN, DCM, FENet, and a paired 3× SRVGG adaptation, plus bicubic. The four-band panel contains RDN, CAFRN, and SRVGG, plus bicubic. The DCM implementation comes from the HSENet comparison repository; the FENet–Land2Sent acronym match has not been verified. These are not a complete reproduction of every model in Land2Sent.

Additional architectures underwent predefined compatibility checks, but only implementations satisfying them entered the quantitative benchmark. The complete status ledger is in Supplementary Table S2. A failed implementation check is not evidence about the architecture's potential accuracy.

EDSR, RCAN, and RDN refer to the published residual-network families (Lim et al., 2017; Zhang et al., 2018b,a); the tested implementations and channel adaptations are defined by the pinned code and per-run constructor receipts, not by their names alone. The TIFFs store signed 16-bit reflectance scaled by 10,000; the evaluator decoded and clipped values to $[0, 1]$ without 8-bit quantization. Inputs were reordered from stored blue–green–red–NIR to model red–green–blue–NIR. All reported test metrics use whole images, raster-validity masks, and a three-pixel border removed on the 480×480 target grid. The release TIFFs have all-valid raster masks and no explicit nodata value; QA/SCL cloud masks were used for pair selection but are not stored as per-pixel scoring exclusions. Thus residual cloud/shadow pixels permitted by the selection threshold remain in the scored images. PSNR and SSIM are arithmetic means of per-image scores, not scores on a pooled mosaic. PSNR uses peak 1.0. For the four-band runs, PSNR, SSIM, MAE, ERGAS, UIQI, and spectral angle use all four channels; the separate RGB PSNR column uses visible channels only. ERGAS uses scale factor three, SAM is reported in degrees, and UIQI is averaged across bands. NDVI is computed from red and NIR. LPIPS uses only the RGB channels with AlexNet features, so it is not a four-band perceptual measure. Times are synchronized single-image model-forward

Table 6: RGB results on all 731 reflectance16 test pairs. Higher PSNR, SSIM, Edge-F1, and UIQI are better; lower MAE, LPIPS, ERGAS, and SAM are better. SAM is in degrees. SRVGG is pixel-loss-only in this run.

Model	PSNR	SSIM	MAE	Edge-F1	LPIPS	ERGAS	UIQI	SAM
RDN	37.67	0.9504	0.0112	0.777	0.128	5.60	0.893	2.92
CAFRN	36.92	0.9399	0.0121	0.694	0.175	6.11	0.870	3.28
DCM	36.80	0.9387	0.0126	0.705	0.170	6.31	0.866	3.64
EDSR	36.76	0.9393	0.0126	0.698	0.169	6.21	0.868	3.46
FENet	36.64	0.9367	0.0128	0.695	0.178	6.45	0.862	3.67
RCAN	36.59	0.9387	0.0129	0.721	0.168	6.39	0.873	3.49
SRVGG (pixel-only)	35.64	0.9108	0.0139	0.649	0.301	7.20	0.804	3.83
Bicubic	35.38	0.9218	0.0144	0.560	0.267	7.32	0.817	4.10

Table 7: Four-band RGB-NIR results on the same 731 test pairs. PSNR-4, SSIM-4, MAE-4, ERGAS, UIQI, and SAM cover RGB and NIR; RGB PSNR is listed separately. Lower MAE, ERGAS, and SAM (degrees) are better; higher values of the other measures are better.

Model	PSNR-4	SSIM-4	MAE-4	ERGAS	UIQI	SAM	RGB PSNR
RDN	36.15	0.9409	0.0126	4.98	0.897	2.73	38.02
CAFRN	34.86	0.9250	0.0148	5.89	0.861	3.22	36.67
SRVGG (pixel-only)	33.91	0.8957	0.0162	6.56	0.814	3.60	35.65
Bicubic	33.89	0.9107	0.0164	6.63	0.828	3.65	35.38

measurements on one NVIDIA A100-SXM4-40GB using float32; they exclude raster decoding, metric computation, and file I/O. Parameter counts are model parameters, not necessarily the number activated for every pixel.

5.2 Quantitative results

RDN has the highest test PSNR in both evaluated panels: 37.67 dB on RGB versus 35.38 dB for bicubic, and 36.15 dB on four bands versus 33.89 dB. Paired mean PSNR gains are 2.291 dB and 2.259 dB. The corresponding descriptive 95% spatial-block bootstrap intervals are [1.601, 3.102] and [1.702, 2.860] dB, respectively: 2,000 draws resample the 13 held-out test blocks with replacement and pool the selected blocks' paired per-image differences. These intervals quantify geography-sensitive variation under the frozen single-seed protocol, not variation across independently trained models. Neither the 731 patches nor the 13 blocks should be read as independent source acquisitions, given the documented product-ID overlap. RGB RDN also improves Edge-F1 from 0.560 to 0.777 and reduces LPIPS from 0.267 to 0.128. Four-band RDN reduces NDVI RMSE from 0.0614 to 0.0503 but its mean NDVI correlation is 0.931, below bicubic's 0.940. Thus improved index error does not imply improvement on every vegetation-index statistic. RGB and four-band PSNR should not be directly ranked against each other because the target channel sets differ.

The SRVGG architecture is listed for completeness, but both SRVGG training runs stopped at epoch 5. The configured perceptual/adversarial phase would only begin after epoch 16; their training logs record zero perceptual and adversarial loss throughout. These checkpoints are therefore *pixel-loss-only SRVGG*, not a trained Real-ESRGAN or a valid GAN-versus-CNN comparison. No separate normalized/quantized-8 training trial is included; the cross-dataset audit below uses only the reflectance16-trained RGB checkpoints.

Footprint-overlap sensitivity. We excluded the 22 test patches with audited positive-area intersection with training or validation footprints and recomputed the archived per-image scores without changing

Table 8: Additional four-band spectral and vegetation-index results on the 731 test pairs. Lower NDVI MAE and RMSE, but higher NDVI correlation r, are better. RGB SSIM uses only visible channels.

Model	NIR PSNR	RGB SSIM	NDVI MAE	NDVI RMSE	NDVI r
RDN	33.49	0.9513	0.0367	0.0503	0.931
CAFRN	32.42	0.9356	0.0429	0.0587	0.917
SRVGG (pixel-only)	31.37	0.9109	0.0500	0.0652	0.919
Bicubic	31.79	0.9218	0.0470	0.0614	0.940

Table 9: RGB PSNR (dB) by test landscape stratum for bicubic and RDN. Counts are patch counts, not independent spatial blocks. Rare strata are descriptive only.

Landscape stratum	n	Bicubic	RDN	Δ PSNR
alpine	101	31.96	32.53	+0.57
cropland	136	37.50	39.69	+2.19
desert/bare	5	29.79	35.34	+5.55
forest	77	37.46	39.76	+2.30
mixed	19	34.87	37.87	+3.00
mountain	170	34.59	37.00	+2.40
shrub/grass	97	35.56	38.75	+3.19
urban	119	35.74	38.35	+2.61
water/wetland	7	36.32	40.52	+4.20

checkpoints, model selection, or the release split. The retained 709 patches still cover all nine strata and all 13 test blocks. RDN remains the highest-PSNR model: RGB PSNR is 37.640 dB versus 35.374 dB for bicubic, a paired gain of 2.266 dB (95% block interval [1.575, 3.078]); four-band PSNR is 36.147 versus 33.890 dB, a gain of 2.257 dB [1.685, 2.875]. Thus the aggregate advantage is not driven by those 22 overlapping patches. This is a post hoc sensitivity analysis, not a replacement benchmark or proof of source independence: 369 retained patches still share an exact acquisition pair with training. Supplementary Section S6 reports every completed model and the exclusion protocol.

Identified-source-fill sensitivity. Original band-window and validity checks distinguish clipped dark values from identified fill in 404 pairs (266 train, 58 validation, 80 test). The affected locations total 148 Landsat pixels and 4,540 Sentinel pixels across the screened rasters; these are not cloud-mask counts. We conservatively exclude the 80 affected test pairs and reaggregate the unchanged archived scores on 651 pairs, retaining all nine strata and 13 blocks. RDN remains highest in PSNR, with gains over same-cohort bicubic of 2.289 dB RGB and 2.237 dB four-band. Joint exclusion of identified-fill pairs and footprint overlaps leaves 629 pairs and gains of 2.260 and 2.234 dB. These are whole-pair exclusion analyses, not pixel-remasked evaluation or corrected training. Their absolute scores reflect a changed scene mix; they do not isolate the causal effect of a few fill pixels. The original 731-pair tables remain the frozen benchmark (Supplementary S9).

5.3 Landscape variation and qualitative examples

The test distribution includes 170 mountain, 136 cropland, 119 urban, and 101 alpine pairs, but only five desert/bare pairs from one block and seven water/wetland pairs from two blocks. Classwise differences therefore describe this particular locked split; the rare-class averages cannot support stable generalization claims. Figures 6–8 show three visually selected test pairs: an urban scene in Punjab, a cropland-stratum scene in Bihar, and a mountain-stratum scene in Jammu and Kashmir. All three use candidate 1 from the frozen full-paper gallery. The examples were selected for landscape coverage and readable features, not

to maximize any model's score; all quantitative comparisons use the complete 731-pair test set. Their full pair IDs and image hashes are in the figure-provenance JSON, and their dates and within-pair time gaps are in Supplementary Section S5. The short-overview gallery uses different samples and is not evidence for these three figures.

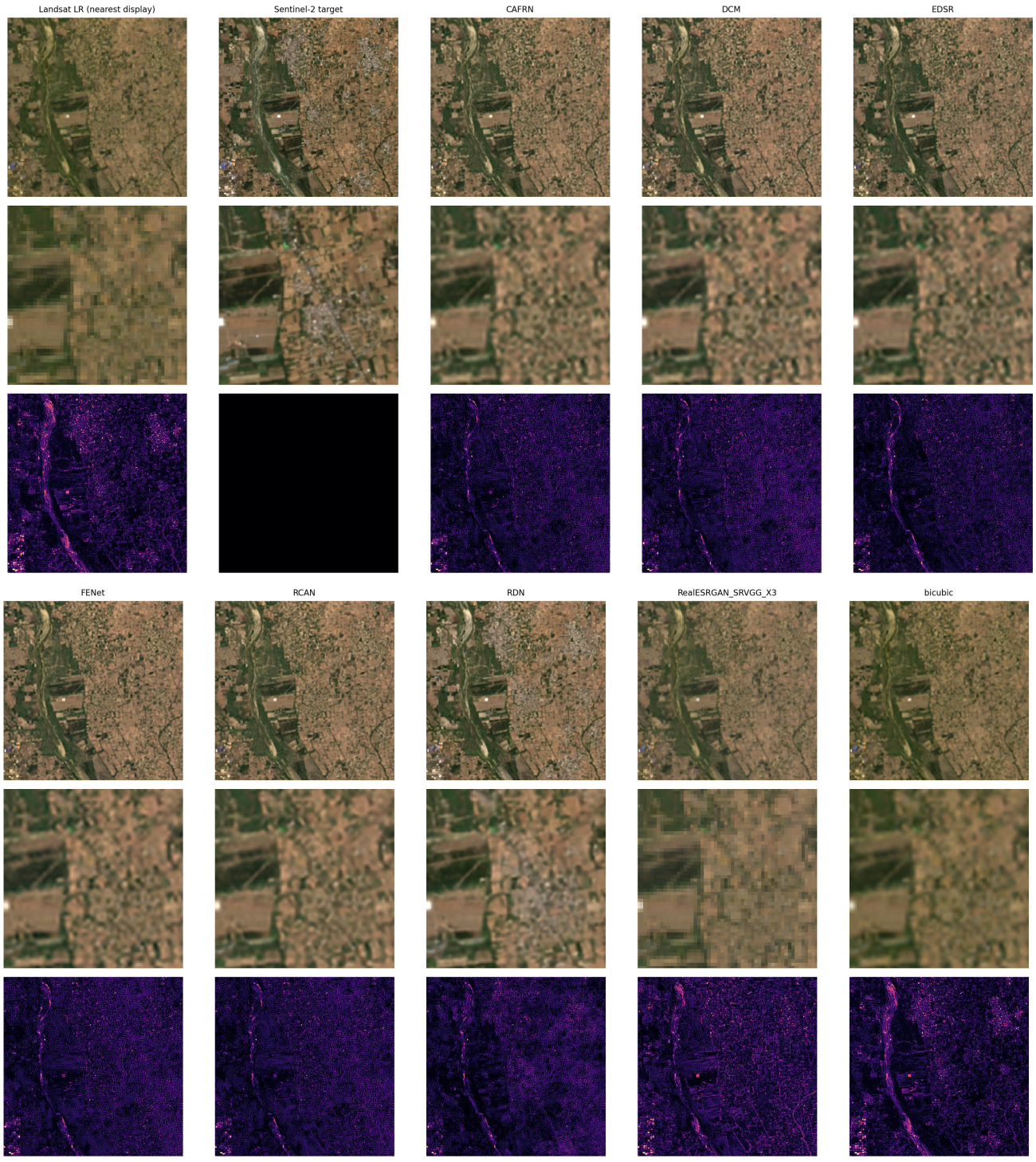


Figure 6: Urban-stratum test pair from Punjab, 16 May 2026 (13.0-minute within-pair gap). The upper strip shows Landsat LR, Sentinel-2 target, CAFRN, DCM, and EDSR; the lower strip shows FENet, RCAN, RDN, pixel-only SRVGG, and bicubic. Rows show the full image, centre zoom, and mean absolute RGB error against Sentinel-2. All RGB panels share a $[0, 0.30]$ reflectance display stretch; error maps share a $[0, 0.10]$ reflectance scale. Neither display scale affects metrics. Full pair ID is in the figure provenance; both source product IDs are in the release manifest.

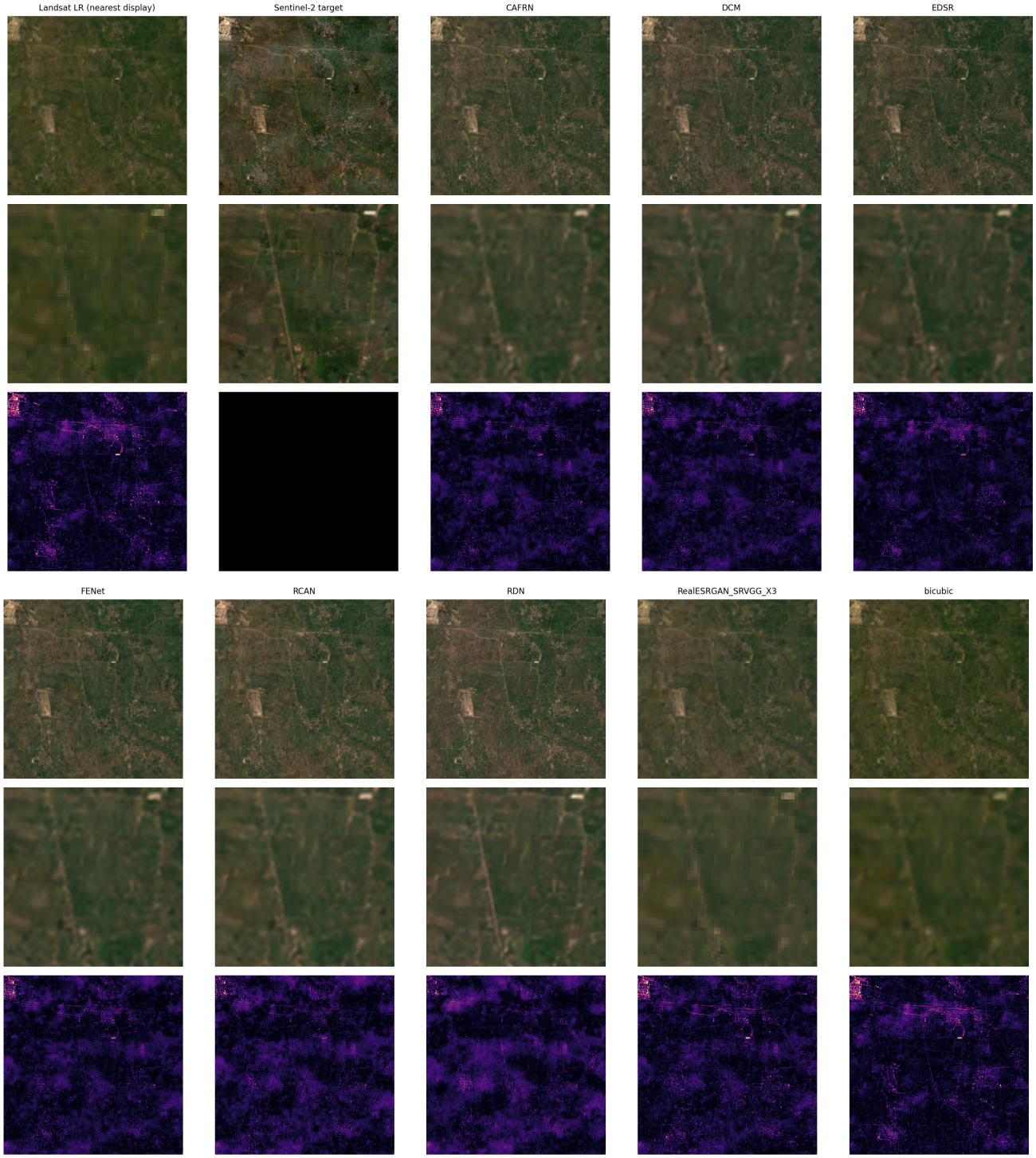


Figure 7: Cropland-stratum test pair from Bihar, 14 January 2026 (12.8-minute within-pair gap), with the same columns and shared $[0, 0.30]$ RGB and $[0, 0.10]$ absolute-error reflectance scales as Figure 6. The landscape label is a sampling stratum, not a claim that every pixel is cropland. Full pair ID is in the provenance record.

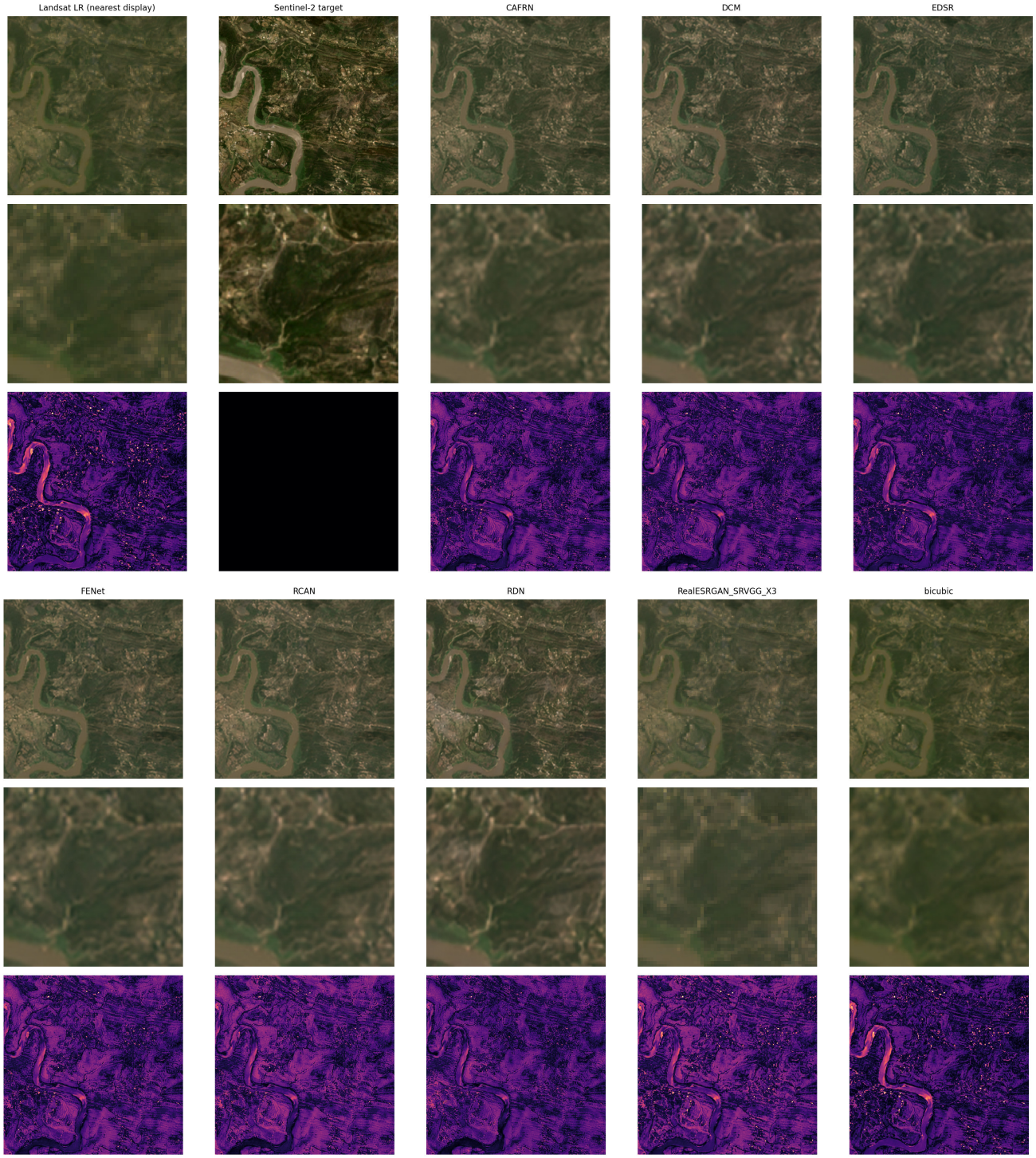


Figure 8: Mountain-stratum test pair from Jammu and Kashmir, 14 May 2026 (10.8-minute within-pair gap), with the same columns and shared $[0, 0.30]$ RGB and $[0, 0.10]$ absolute-error reflectance scales as Figure 6. The visible river valley is part of a DSM-labelled mountain-region footprint. Full pair ID is in the provenance record.

5.4 Additional diagnostics and efficiency

Tables 10–12 report the remaining full-test diagnostics retained by the evaluator. Y-channel scores are defined on the RGB projection; RGB LPIPS likewise uses only RGB for both profiles. The per-band PSNRs should not be substituted for the four-band PSNR, which is computed from the multichannel error for each image and then averaged. The operation counts use the harness's `fvcore` forward-operator convention on a 160×160 LR input, so they are indicative rather than hardware-independent wall-clock predictions.

Table 10: Additional spatial and perceptual diagnostics for all 731 test pairs per run. Higher Y PSNR, Y SSIM, Edge-F1, edge PSNR, gradient correlation, and sCC are better; lower RMSE and RGB LPIPS are better. All values are means of per-image metrics.

Profile	Model	RMSE	Y PSNR	Y SSIM	Edge-F1	Edge PSNR	Grad. r	sCC	RGB LPIPS
RGB	RDN	0.0147	38.16	0.9562	0.777	47.28	0.850	0.669	0.128
RGB	CAFRN	0.0159	37.52	0.9473	0.694	46.29	0.812	0.614	0.175
RGB	DCM	0.0165	37.43	0.9469	0.705	46.25	0.813	0.620	0.170
RGB	EDSR	0.0166	37.37	0.9464	0.698	46.29	0.813	0.621	0.169
RGB	FENet	0.0165	37.30	0.9456	0.695	46.15	0.803	0.594	0.178
RGB	RCAN	0.0166	37.21	0.9471	0.721	46.43	0.817	0.616	0.168
RGB	SRVGG (pixel-only)	0.0183	36.22	0.9200	0.649	43.78	0.698	0.380	0.301
RGB	Bicubic	0.0190	36.27	0.9325	0.560	44.29	0.773	0.597	0.267
RGBNIR	RDN	0.0171	38.54	0.9574	0.772	47.32	0.849	0.710	0.129
RGBNIR	CAFRN	0.0200	37.38	0.9448	0.674	45.94	0.802	0.667	0.181
RGBNIR	SRVGG (pixel-only)	0.0219	36.22	0.9199	0.648	43.78	0.697	0.449	0.301
RGBNIR	Bicubic	0.0222	36.27	0.9325	0.560	44.29	0.773	0.662	0.267

Table 11: Per-band test PSNR (dB), computed before averaging across 731 images. RGB-only runs have no NIR score.

Profile	Model	Red PSNR	Green PSNR	Blue PSNR	NIR PSNR
RGB	RDN	36.83	38.41	38.32	–
RGB	CAFRN	36.26	37.80	37.27	–
RGB	DCM	36.20	37.73	37.01	–
RGB	EDSR	36.04	37.66	37.19	–
RGB	FENet	36.04	37.61	36.86	–
RGB	RCAN	35.89	37.51	36.98	–
RGB	SRVGG (pixel-only)	34.72	36.64	36.23	–
RGB	Bicubic	34.74	36.90	35.36	–
RGBNIR	RDN	37.39	38.70	38.46	33.49
RGBNIR	CAFRN	36.12	37.62	36.85	32.42
RGBNIR	SRVGG (pixel-only)	34.72	36.63	36.25	31.37
RGBNIR	Bicubic	34.74	36.90	35.36	31.79

Table 12: Model size, measured inference latency, and completed training epochs. "G ops" is the `fvcore` operator count in billions for one 160×160 input; bicubic has no network parameter or operator count. Times are synchronized A100 model-forward means and 95th percentiles (ms), excluding I/O and metrics.

Profile	Model	M params	G ops	Mean ms	P95 ms	Epochs
RGB	RDN	22.31	571.2	38.0	42.1	22
RGB	CAFRN	1.14	575.9	57.5	57.6	14
RGB	DCM	2.26	57.8	6.7	6.7	16
RGB	EDSR	43.68	1119.1	41.6	53.4	10
RGB	FENet	0.36	8.7	10.6	10.7	10
RGB	RCAN	15.63	397.1	51.9	52.3	14
RGB	SRVGG (pixel-only)	1.20	30.6	4.3	4.3	5
RGB	Bicubic	0.00	–	0.1	0.1	–
RGBNIR	RDN	22.31	571.3	57.6	57.8	30
RGBNIR	CAFRN	1.15	576.1	70.5	72.1	7
RGBNIR	SRVGG (pixel-only)	1.21	30.8	4.3	4.3	5
RGBNIR	Bicubic	0.00	–	0.1	0.1	–

6 Recommended Use, Limitations, and Reproducibility

6.1 Recommended evaluation protocol

The provided split is disjoint by 100-km block ID, but it is neither source-product-disjoint nor guaranteed to have a nonzero intersplit footprint gap. Users should not randomly reshuffle individual patches, because this can place nearby observations or patches from the same source scenes in both training and test sets. A study of generalization to unseen acquisitions or buffered geography must construct and version an appropriate new partition, then retrain and retest all models. Model selection belongs on validation data; test data should remain unopened until the experimental protocol is frozen. Metrics should be computed on physical reflectance in $[0, 1]$, not gamma-corrected display images. RGB-only work should explicitly select red, green, and blue while preserving the release's band order (blue, green, red, NIR). Multispectral work should report both four-band and RGB metrics and should include a vegetation-index error such as NDVI MAE.

Because the mapping is cross-sensor, a complete evaluation should report more than PSNR alone. The present benchmark reports L1, PSNR, SSIM, SAM, ERGAS, UIQI, spatial correlation, and edge-sensitive scores; four-band runs also report NDVI error. LR re-degradation consistency would require a calibrated sensor-specific degradation operator and is *not* measured here. Statistical comparisons should use paired per-image differences and spatial-block confidence intervals. Any crop-based protocol must state crop size, crop location or randomization, number of crops per source, and whether multiple crops from one source are treated as independent observations.

6.2 Cross-dataset transfer audit

We applied the frozen Mahimandal-trained RGB checkpoints, without fine-tuning or test-driven selection, to all 100 official OLI2MSI test pairs. The OLI2MSI TIFFs are three-band float32 reflectance images with 160×160 Landsat inputs and 480×480 Sentinel-2 references; paired NIR is unavailable. The *reflectance16* label below identifies the training-release encoding, not the OLI2MSI TIFF dtype. Evaluation used the same three-pixel border exclusion and RGB channel metrics as the source benchmark. Table 13 compares each checkpoint with a bicubic prediction on the *same* OLI2MSI pairs; source and target PSNR levels should not be directly subtracted to infer a domain effect because scene content and acquisition conditions differ.

Only FENet exceeds the OLI2MSI bicubic reference in PSNR, by 0.193 dB; it also has the highest transferred SSIM and Edge-F1 in this panel. The remaining learned models are below bicubic in PSNR, including the Mahimandal-leading RDN. This demonstrates that high in-domain rank does not guarantee zero-shot cross-dataset rank. It does not establish why transfer fails: scene distribution, sensor processing, radiometry, and source-pair construction may all contribute. Neither OLI2MSI training nor reverse transfer is represented by this table. The DCM provenance and FENet identity caveats stated above still apply. The figures and per-pair CSV in the Notebook-3 result bundle permit reanalysis; this table reports only the fixed benchmark baselines.

The transferred OLI2MSI loader reads bands 1–3 as RGB, checks paired $160 \times 160 \rightarrow 480 \times 480$ geometry and valid-pixel masks, and clips the supplied float32 values to $[0, 1]$ without applying the Mahimandal 16-bit scale. A subsequent all-file audit supports the RGB interpretation through TIFF color tags and the publisher's extraction code, despite absent band descriptions. The downloaded paired-score table gives FENet's $+0.193$ dB gain over same-pair bicubic a 95% interval of $[0.093, 0.277]$ dB and RDN's -2.166 dB difference an interval of $[-2.307, -2.030]$ dB. These are 2,000-draw paired *image*-bootstrap intervals with seed 42, not geographic-block intervals; OLI2MSI test pairs may share source scenes. They do not isolate a cause of the cross-dataset rank change.

The local audit additionally inspects all 200 OLI2MSI test TIFFs: each is three-band float32 with RGB interpretation tags, finite values within $[0, 1]$, and the archived test-pair identifiers. This supports the loader's channel mapping despite absent band descriptions. The publisher's pinned extraction code selects

Table 13: Zero-shot transfer of Mahimandal-trained RGB checkpoints to the 100-pair OLI2MSI test split. ΔPSNR is relative to OLI2MSI bicubic (42.215 dB), not to Mahimandal test PSNR. All numbers are descriptive single-checkpoint test means.

Model	Mah. PSNR	OLI PSNR	ΔPSNR	OLI SSIM	OLI Edge-F1
Bicubic	35.378	42.215	0.000	0.9644	0.6021
EDSR	36.763	39.258	-2.957	0.9613	0.5103
RCAN	36.595	36.869	-5.346	0.9506	0.5884
RDN	37.669	40.048	-2.166	0.9630	0.5067
CAFRN	36.922	35.991	-6.223	0.9481	0.5308
DCM	36.796	31.258	-10.957	0.9135	0.5186
FENet	36.644	42.407	+0.193	0.9660	0.6444
SRVGG (pixel-only)	35.644	38.398	-3.816	0.9529	0.5730

bands 4/3/2 and invokes top-of-atmosphere reflectance readers for both sensors (Wang et al., 2021b); Mahimandal uses surface reflectance. A shared numeric range is therefore not atmospheric harmonization. All 100 local pairs have different LR/HR header bounds in the same CRS; these coordinate differences are not image-registration residuals. We preserve the supplied paired arrays and do not rewarp the official test images. Publisher code and TIFF tags do not authenticate every repackaged pixel against an original signed archive. These distinctions further limit causal interpretations of transfer (Supplementary S10).

Architecture contributions, including any companion super-resolution method developed by the authors, are outside this article and are not used as evidence for the dataset contribution.

6.3 Known limitations

- **Cross-sensor target mismatch.** Sentinel-2 differs from Landsat in spectral response, spatial response, viewing geometry, and atmospheric processing. The target is a reference, not exact 10-m Landsat truth.
- **Annual labels are sampling metadata.** Esri 2025 land cover is contemporaneous at annual scale for 2025 acquisitions, but not exact-date ground truth; it is one year older for the 2026 majority. ESA WorldCover 2021 is historical comparison only. Neither should be used as a supervised semantic target without independent validation.
- **Terrain is a DSM.** Copernicus DEM GLO-30 includes buildings and vegetation. Elevation and slope are useful sampling descriptors, not surveyed terrain measurements.
- **Temporal imbalance.** The dataset is dominated by 2026 and January acquisitions. It does not represent seasons uniformly.
- **Rare-class shortfalls.** Water/wetland, desert/bare, and mixed strata remain below their requested reserves. Results for those strata will have wider uncertainty.
- **Residual alignment proxy.** Phase correlation summarizes residual image displacement but is not a replacement for independently surveyed control points or a full local deformation model.
- **Block and source overlap.** Zero 100-km block IDs are shared across splits, but 22 test patches have audited positive-area footprint overlap with training or validation. Excluding them leaves 709 patches in the same 13 blocks, including 369 patches sharing a source-product pair with training. The sensitivity check does not establish a spatial buffer, source-disjointness, or independent geographic replicates.
- **Geographic labels.** The openly licensed boundary layer is reproducible but not an official national boundary. Two pair centres lack a usable ADM1 name in the current metadata.

- **Selection bias.** Strict clear-sky, texture, correlation, overlap, land-cover availability, and quota filters favor scenes that satisfy the construction protocol. The rejection log must accompany the dataset so this bias remains observable.

6.4 Reproducibility and release controls

Release v1 includes the construction and audit code, the immutable manifest, pair-level metric cache, rejection counts, split files, report tables, figure provenance, a machine-readable data card, protocol metadata, a data-license and source-attribution notice, and SHA-256 inventories covering packaged imagery and metadata. The two storage variants contain identical pairs and split memberships: one preserves signed 16-bit reflectance multiplied by 10,000, and one stores decoded float32 reflectance in $[0, 1]$. They are storage encodings, not independent datasets or separate benchmark trials. The release records the minimum 11% validation and test holdouts, the realized 71.33/13.44/15.23 percentages, and a zero-missing/zero-orphan closure audit. Benchmark table sources, the manifest hash, protocol files, and checkpoint receipts are included in the associated result bundle. Any change to samples or assignments requires a new version, manifest hash, checksums, statistics, and figures.

Original curation, manifest, and documentation are licensed under CC BY 4.0. That license does not replace source terms: Landsat observations remain public-domain USGS data; Sentinel-2 and Copernicus products retain their applicable data terms; and auxiliary products retain their respective licenses. The release notice records these attributions rather than asserting new rights over source observations.

No imagery was collected from private systems and no person-level labels are provided. Nevertheless, 10-m imagery can expose infrastructure and settlement patterns. Users should follow applicable geospatial-data policy, avoid harmful targeting applications, and interpret administrative boundaries with care.

7 Conclusion

MahimandalSR provides 4,800 real, same-day Landsat–Sentinel-2 RGB-NIR pairs for 3× cross-sensor super-resolution across Bharat. Its distinguishing features are minute-scale acquisition separation, explicit patch-level quality control, Himalayan and landscape-aware sampling, physical-reflectance GeoTIFFs, and spatial-block evaluation. On the frozen 731-patch test split spanning 13 held-out blocks, the strongest evaluated baseline, RDN, exceeds bicubic by 2.29 dB on RGB and 2.26 dB on four bands, demonstrating a measurable reconstruction task under the specified protocol. In a separate 100-pair OLI2MSI test, most Mahimandal-trained RGB checkpoints fall below bicubic PSNR; this is evidence of limited zero-shot transfer, not of OLI2MSI-trained performance. These are single-seed results with limited model coverage and shared source acquisitions across the Mahimandal splits. Annual land-cover labels are metadata, residual shifts are proxies, class and season balance are imperfect, and blockwise splits must remain intact. Release v1 excludes non-manifest staging files and provides machine-readable closure and checksum records.

Data and Code Availability

For confidential peer review, the frozen reflectance16 release, its normalized-float32 storage variant, the release manifest and checksums, the source and analysis code, and the benchmark result bundle will be provided to editors and reviewers through private access details supplied with the submission. The review package is identified by the reflectance16 manifest SHA-256 (prefix `6bf97403ee40ccf0`; full digest in Supplementary Section S1); both storage variants preserve the same 4,800 pair IDs and split assignments. After publication, the variants will be made public as the reflectance16 dataset and normalized-float32 dataset, with versioned code, construction scripts, evaluation protocols, and result artifacts in the source repository. Source imagery remains subject to the original providers' terms; the release documentation distinguishes these from the CC BY 4.0 licence on original curation and metadata.

Acknowledgements

Landsat imagery is provided by the U.S. Geological Survey. Sentinel-2 and Copernicus products, including Copernicus DEM GLO-30, are provided under the Copernicus programme. The land-cover sampling metadata were produced by Impact Observatory, Microsoft, and Esri. Administrative labels use geoBoundaries gbOpen, and ESA WorldCover is acknowledged as the historical comparison source.

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